

Process Transfer Plan

**A-Mab Drug Substance — Cambridge (Development) → Grafton (Commercial DS) —
PTP-001**

Manufacturing Science & Technology (MSAT) / Technology Transfer

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Technology Transfer / MSAT	—	—	—
Reviewer	Sending site — Process Development	—	—	—
Reviewer	Receiving site — Manufacturing	—	—	—
Reviewer	Analytical Development / QC	—	—	—
Approver	Quality Assurance (both sites)	—	—	—

Abbreviations

AMV, analytical-method validation · BLA, Biologics License Application · CPP, critical process parameter · CQA, critical quality attribute · DP, drug product · DS, drug substance · GPP, general process parameter · ICH, International Council for Harmonisation · IPC, in-process control · KPP, key process parameter · LRV, log-reduction value · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PAR, proven acceptable range · PC, process characterization · PPQ, process performance qualification · QbD, Quality by Design · SDM, scale-down model · WC-CPP, well-controlled CPP.

1 Purpose and scope

This Process Transfer Plan (**PTP-001**) defines the strategy and scope for transferring the A-Mab drug-substance (DS) manufacturing process from the **sending site** (Novacyte Biologics — Cambridge, MA (Development)) to the **receiving site** (Novacyte Biologics — Grafton, WI (Commercial DS)). It transfers the process knowledge developed and characterized during Stage 1 (Process Design) — the parameter classifications, proven acceptable ranges, multivariate design spaces and per-CQA control strategy captured in the process-characterization package — to the commercial DS facility, and defines the site-equivalency analysis, the scale-down-model and comparability strategy, the transfer of the manufacturing process, analytical methods and control strategy, the PPQ / batch strategy, and the gap analysis that governs execution.

Scope. The 8-step commercial-scale A-Mab DS process (production bioreactor through final UF/DF) at the 15,000-L commercial scale, including the transfer of the validated analytical methods and the control strategy. Drug- product (fill/finish) transfer and the formulation characterization are addressed under the drug-product program and are out of scope. This plan governs the transfer strategy; the receiving-site validation (PPQ) is executed under the site validation plan and confirmed in the Process Characterization Master Report (**PCMR-001**).

Basis. The transfer follows the technology-transfer good practices of the ISPE Technology Transfer guide (International Society for Pharmaceutical Engineering 2023b) and the lifecycle transfer strategy of PDA TR 60 §6 and the ISPE PV guide (Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023a), within the QbD framework of ICH Q8, Q9, Q10 and Q11 (International Council for Harmonisation 2009, 2023, 2008, 2012) and the process model and control-strategy framework of the A-Mab case study (CMC Biotech Working Group 2009). The process knowledge transferred is that established in the per-unit-operation characterization plans and reports and consolidated in the Process Characterization Master Plan (**PCMP-001**) and Master Report (**PCMR-001**).

2 Related documents

This plan is the parent transfer document for the A-Mab DS process-characterization package; the full package is cross-referenced for traceability.

Table 2.1: The A-Mab drug-substance document package referenced by this transfer plan.

Document ID	Document class	Subject
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

3 Transfer strategy

The transfer applies the enhanced (QbD) approach: because the process was developed and characterized against a defined set of CQAs with multivariate design spaces and a per-CQA control strategy, the transfer is a **knowledge transfer** of that understanding to the receiving site, confirmed by site-equivalency analysis, engineering runs and a commercial-scale PPQ campaign rather than by re-development. The strategy comprises, consistent with PDA TR 60 §6 (Parenteral Drug Association 2013): (i) a product and process description carried from the characterization package; (ii) a site-equivalency analysis comparing the sending and receiving equipment, facilities and utilities; (iii) a scale-down-model and comparability strategy that bridges development and commercial scale; (iv) transfer of the manufacturing process, the analytical methods and the control strategy; (v) a PPQ / batch strategy that confirms the CPPs, the design spaces and the cumulative viral clearance at commercial scale; and (vi) a gap analysis with mitigations. The criticality analysis and risk assessments are confirmed at the receiving site (**RA-001**, **PCMP-001**).

4 Product and process description

A-Mab is a humanized IgG1 monoclonal antibody produced by fed-batch cell culture and purified by a platform train. The commercial process operates at 15,000 L with a nominal titre of 4.37 g/L and an overall DS yield of 83.2%, delivering approximately 54.6 kg of drug substance at 75 g/L. The 8 unit operations, and each step’s principal role in the control strategy, are summarized in Table 4.1.

Table 4.1: The A-Mab drug-substance process train (Steps 3–10) and each step’s principal role in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The process controls 10 drug-substance CQAs. Their acceptance criteria and criticality, established in the characterization package, are the quality targets the transfer must preserve; the integrated control strategy delivers them with a minimum commercial-scale capability of $C_{pk} = 1.51$ and cumulative viral clearance of 18.87 log (XMuLV) and 10.03 log (MVM).

Table 4.2: Drug-substance CQAs, acceptance criteria and criticality (Tool #1 = Impact $\times$ Uncertainty) preserved across the transfer.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48

CQA	Category	Acceptance	Criticality	Tool #1
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20

5 Sending and receiving sites

The transfer moves the process from the development site, where it was characterized on qualified scale-down models, to the commercial DS facility (Table 5.1).

Table 5.1: Sending and receiving sites for the A-Mab drug-substance transfer.

Role	Site	Location	Function in the transfer
Sending site	Novacyte Biologics — Cambridge, MA (Development)	Cambridge, MA	Development; Stage-1 process characterization on scale-down models
Receiving site	Novacyte Biologics — Grafton, WI (Commercial DS)	Grafton, WI	Commercial drug-substance manufacture; Stage-2 PPQ

6 Site-equivalency analysis

A site-equivalency analysis compares the receiving-site equipment train, facilities and utilities against the sending-site (and scale-down) reference to assure they are equivalent and qualified for commercial manufacture (Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023b). The analysis covers, for each unit operation: the equipment make, operating principle and qualification status; the operating ranges for the CPPs and WC-CPPs at the new scale; the buffer, media and consumable supply; the automation and control system; and the personnel qualification and training. Any non-equivalence is captured as a transfer gap (§11) with a mitigation and is resolved before the PPQ campaign. Equivalency is confirmed by engineering runs at commercial scale prior to PPQ.

7 Scale-down model and comparability strategy

The commercial-scale process is bridged to the characterization data by the **qualified scale-down models** on which the design spaces and control strategy were established. The comparability strategy compares, between the scale-down model and commercial scale: the step yields; the CQA outcomes (glycans, charge variants, aggregate, HCP, residual DNA, leached Protein A); and the modular viral clearance. A comparability plan describes the actions to be taken in the event of a significant process difference at scale, including the testing program used to demonstrate comparability between the characterized and the commercial processes (Parenteral Drug Association 2013). Confirmation at commercial scale is provided by the PPQ campaign and reported in **PCMR-001**.

8 Transfer of the manufacturing process

The manufacturing process is transferred by carrying the characterized set-points, normal operating ranges (NORs) and proven acceptable ranges (PARs) into the receiving-site master batch records, and by confirming the CPP and WC-CPP ranges at commercial scale in engineering and PPQ batches. Of the 37 process parameters characterized across the train, 21 are critical (1 CPP and 20 WC-CPP), and are the parameters whose commercial-scale control the transfer must confirm; the remaining 10 KPP and 6 GPP are transferred as process-performance and general controls. The critical parameters and their NORs are summarized in Table 8.1.

Table 8.1: Critical process parameters (CPP / WC-CPP) and their normal operating ranges, confirmed at commercial scale during the transfer.

Unit operation	Parameter	Unit	Set-point	NOR	Class
Production Bioreactor	Culture pH	pH	6.85	6.75–6.95	WC-CPP
Production Bioreactor	Culture temperature	°C	35	34.5–35.5	WC-CPP
Production Bioreactor	Dissolved CO ₂ (pCO ₂)	mmHg	100	70–130	WC-CPP
Production Bioreactor	Osmolality	mOsm	400	375–425	WC-CPP
Production Bioreactor	Culture duration	days	17	16–18	WC-CPP
Protein A Chromatography	Protein load	g/L resin	30	20–40	WC-CPP
Protein A Chromatography	Elution buffer pH	pH	3.55	3.4–3.7	WC-CPP
Low-pH Viral Inactivation	Inactivation pH	pH	3.5	3.4–3.6	CPP
Low-pH Viral Inactivation	Hold time	min	90	60–120	WC-CPP
Low-pH Viral Inactivation	Temperature	°C	21	19–23	WC-CPP
Cation Exchange Chromatography	Protein load	g/L resin	25	18–32	WC-CPP
Cation Exchange Chromatography	Load/Wash conductivity	mS/cm	5	4–6	WC-CPP
Cation Exchange Chromatography	Elution buffer pH	pH	6	5.85–6.15	WC-CPP
Cation Exchange Chromatography	Elution stop collect	OD desc.	1	0.7–1.3	WC-CPP

Unit operation	Parameter	Unit	Set-point	NOR	Class
Anion Exchange Chromatography	Load pH	pH	7.5	7.35–7.65	WC-CPP
Anion Exchange Chromatography	Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	WC-CPP
Anion Exchange Chromatography	Load conductivity	mS/cm	5.5	4–7	WC-CPP
Anion Exchange Chromatography	Protein load	g/L resin	200	120–280	WC-CPP
Anion Exchange Chromatography	Operating flow rate	cm/hr	300	200–400	WC-CPP
Small Virus Retentive Filtration	Filtration volume (load)	L/m ²	90	50–105	WC-CPP
Small Virus Retentive Filtration	Filtration pressure	psi	13	10–20	WC-CPP

9 Transfer of analytical methods

The validated release, in-process and characterization methods are transferred to the receiving-site QC laboratory under method-transfer protocols with cross-site precision and accuracy acceptance criteria, and co-validated before the PPQ campaign. The methods include the glycan map, size-variant (SEC), charge-variant (icIEF), HCP, residual-DNA, leached-Protein-A and the XMuLV/MVM infectivity assays that measure the CQAs across the train. Method equivalence between sites is a prerequisite for the comparability and PPQ assessments; any method gap is captured in §11.

10 Transfer of the control strategy

The per-CQA control strategy defined during characterization — input-material controls, procedural controls, process-parameter controls (CPP/WC-CPP NORs), in-process controls, and release and characterization testing — is transferred by mapping every control element into the receiving-site master batch records and in-process-control plan, and confirmed in PPQ. The control strategy assures the 10 CQAs at commercial scale, with the tightest capability at the cumulative viral-clearance CQA ($C_{pk} = 1.51$); the one true CPP of the process (the low-pH viral-inactivation pH) and the 20 WC-CPPs receive enhanced control. The overall control strategy is consolidated in **PCMR-001**.

11 PPQ and batch strategy

Commercial-scale confirmation is provided by a Stage-2 process performance qualification (PPQ) campaign at the receiving site. The PPQ batches reproduce the manufacturing process, components and composition at commercial scale and confirm the CPP ranges, the multivariate design spaces and the cumulative viral clearance (Parenteral Drug Association 2013). The number of PPQ batches is set by the receiving-site validation plan (the characterization capability model uses 5 qualification batches at NOR-distributed set-points as the basis for the commercial-scale capability estimate); the PPQ acceptance criteria are the CQA acceptance criteria and the confirmed CPP ranges. PPQ results are reported in **PCMR-001**.

12 Gap analysis

The gap analysis compares the sending and receiving sites and identifies the differences that must be resolved before or during the transfer, each with an impact assessment, a mitigation and a status (Table 12.1). Open gaps are resolved before the PPQ campaign; their closure is tracked in the transfer record.

Table 12.1: Transfer gap analysis (summary; full impact and mitigation in Table 12.2).

Gap	Area	Description	Status
GAP-01	Equipment	The commercial-scale chromatography skids and production bioreactor at the receiving site differ in make and control system from the development equipment.	Open
GAP-02	Analytical method	The validated release and characterization methods must be transferred and co-validated at the receiving-site QC laboratory.	In progress
GAP-03	Facility	The Grafton commercial drug-substance facility and its utilities require qualification for A-Mab manufacture.	Open
GAP-04	Materials	The resin, virus-retentive membrane, depth-filter media and cell-culture media/feed lots at commercial scale differ from the development lots.	Open
GAP-05	Process	The characterized CPPs, WC-CPPs and the multivariate design spaces must be confirmed at commercial scale.	Open
GAP-06	Control strategy	The per-CQA control strategy defined during characterization must be implemented in the receiving-site master batch records and in-process-control plan.	In progress

Table 12.2: Transfer gap analysis — impact and mitigation.

Gap	Impact	Mitigation
GAP-01	Operating ranges and control response may differ at commercial scale.	Engineering runs and equipment qualification confirm that the CPP and WC-CPP ranges transfer; the qualified scale-down models bridge development and commercial scale.
GAP-02	Method bias between sites could confound the comparability assessment.	Method-transfer protocols with cross-site precision and accuracy acceptance criteria are executed before the PPQ campaign.
GAP-03	Facility fit, segregation and environmental controls must be demonstrated for the process.	Facility, utility and equipment qualification is completed before the engineering and PPQ batches.

Gap	Impact	Mitigation
04	GAPRaw-material and consumable lot-to-lot variability could affect clearance, yield or product quality.	Raw-material and consumable lot bridging with incoming specifications and resin/membrane life-cycle control is applied.
05	GAPScale effects on the design space and on the cumulative viral clearance must be verified.	The Stage-2 PPQ campaign confirms the CPP ranges, the design spaces and the cumulative viral clearance at commercial scale.
06	GAPIncomplete transfer of in-process controls and limits could weaken the control of a CQA.	The control strategy is transferred by mapping every in-process control and limit into the master batch records and confirming them in PPQ.

13 Roles and responsibilities

Table 13.1: Roles and responsibilities for the transfer.

Function	Responsibility
Technology Transfer / MSAT	Author and own the transfer plan; coordinate the transfer and gap closure
Sending site — Process Development	Provide the characterized process knowledge, ranges and scale-down-model data
Receiving site — Manufacturing	Execute engineering and PPQ batches; implement the master batch records
Analytical Development / QC	Transfer and co-validate the analytical methods; support comparability
Quality Assurance (both sites)	Approve the plan; confirm the risk assessments and control strategy; release PPQ

14 Schedule

The transfer proceeds in sequence: (1) site-equivalency analysis and facility/equipment qualification; (2) analytical-method transfer and co-validation; (3) engineering runs confirming the CPP ranges and scale-down-model comparability; (4) the Stage-2 PPQ campaign; and (5) the Process Characterization Master Report (**PCMR-001**) confirming the transfer and commercial-scale control strategy. Gap closure (§11) gates entry into the PPQ campaign.

15 Risks and assumptions

The plan assumes the receiving-site facility, utilities and equipment are qualified for A-Mab manufacture and that the analytical methods transfer with acceptable cross-site equivalence. The principal risk is that scale effects shift a CPP range, a design space or the cumulative viral clearance at commercial scale; this is mitigated by the scale-down-model bridge, the site-equivalency analysis, the engineering runs and the PPQ confirmation, with the gap analysis (§11) tracking every non-equivalence to closure before PPQ.

16 Approvals

Approval of this plan authorizes the transfer of the A-Mab drug-substance process described herein. Signatures are recorded in Table [1](#).

References

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